

Machine Learning Unit-wise Exam Notes

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7-Mark Answer Ready Notes

1. Machine Learning Kya Hai?

- Machine Learning ek aisi technique hai jisme computer data se seekh kar future predictions ya decisions leti hai.
- Isme explicit programming ki jagah data par training di jati hai.
- Ye supervised, unsupervised aur reinforcement learning mein classify hoti hai.
- Iska use healthcare, finance, NLP, computer vision aur recommendation systems mein hota hai.
- Iske main advantages hain automation, pattern recognition aur better decision making.
- Limitations mein data quality, overfitting, underfitting aur high computational cost shamil hain.

2. Supervised aur Unsupervised Learning

- Supervised learning mein labeled data use hota hai jahan input aur output dono known hote hain.
- Regression aur classification iske examples hain.
- Unsupervised learning mein labeled data nahi hota aur system data ko groups ya patterns mein organize karta hai.
- Clustering aur PCA iske examples hain.
- Supervised learning prediction ke liye aur unsupervised learning insight ke liye use hoti hai.

3. Regression Kya Hai?

- Regression continuous values ke liye use hoti hai jaise house price ya sales prediction.
- Iska main aim ek relationship establish karna hota hai input variables aur output variable ke beech.
- Linear regression sabse basic model hai jisme straight-line relationship assume ki jati hai.
- Polynomial regression complex relationships ko handle karta hai.
- Regression ko evaluate karne ke liye MSE, RMSE aur MAE use hote hain.

4. Neural Network Kya Hai?

- Neural network biological brain se inspire hokar banaya gaya model hai.
- Isme input layer, hidden layers aur output layer hote hain.
- Each neuron weights aur bias ke through information process karta hai.
- Activation functions jaise Sigmoid, Tanh aur ReLU non-linearity introduce karte hain.
- Neural networks complex patterns samajh sakte hain aur deep learning mein bohot important hain.

5. CNN Kya Hai?

- CNN specially image aur spatial data ke liye design ki gayi deep learning model hai.
- Isme convolution layer features extract karta hai, pooling layer size reduce karta hai aur dense layer final prediction karta hai.
- Flattening 2D feature maps ko 1D vector mein convert karta hai.
- Padding aur stride output size ko control karte hain.
- CNN image classification, object detection aur face recognition mein use hoti hai.

6. RNN, LSTM aur GRU

- RNN sequential data ke liye use hoti hai jahan previous information important hoti hai.

- RNN vanishing gradient problem ka samna karti hai.
- LSTM is problem ko solve karta hai aur long-term dependencies ko remember karta hai.
- GRU LSTM ka simpler aur faster version hai.
- Ye NLP, speech aur time-series data mein use hote hain.

7. Reinforcement Learning Kya Hai?

- Reinforcement learning mein agent environment ke saath interact karta hai aur reward ke basis par seekhta hai.
- Agent ka goal maximum cumulative reward lana hota hai.
- Isme agent, environment, state, action aur reward ka concept important hai.
- MDP decision-making framework provide karta hai.
- Q-learning aur SARSA popular RL algorithms hain.

8. SVM aur Bayesian Learning

- SVM classification aur regression dono ke liye use hoti hai aur maximum margin boundary banati hai.
- Bayesian learning probability-based reasoning par based hoti hai.
- Ye prior aur posterior probability ke concept par kaam karti hai.
- SVM aur Bayesian methods dono strong probabilistic aur mathematical models hain.

Unit I – Basics of Machine Learning

1. Introduction to Machine Learning

- Machine Learning ek aisi technique hai jisme machine data se seekh kar decisions leti hai.
- Iska use prediction, classification, clustering aur pattern recognition mein hota hai.
- Main objective: past data se future outcomes ka estimate karna.

2. Scope

- AI, data science, healthcare, finance, robotics, NLP, computer vision aur recommendation systems mein use hoti hai.
- Scope bahut broad hai.

3. Limitations

- Data quality par depend karti hai.
- Overfitting aur underfitting issues aa sakte hain.
- Interpretability mushkil ho sakti hai.
- Large computational resources chahiye hote hain.

4. Regression

- Regression continuous output ke liye use hoti hai.
- Examples: house price prediction, sales forecasting.
- Common types: Linear Regression, Polynomial Regression.

5. Probability

- Probability events ki likelihood ko describe karta hai.
- Bayes theorem, conditional probability aur random variables important hain.

6. Statistics for Machine Learning

- Mean, median, mode, variance, standard deviation, covariance aur correlation important hain.
- Data ko samajhne aur model evaluate karne ke liye use hoti hai.

7. Linear Algebra for Machine Learning

- Vectors, matrices, eigenvalues, eigenvectors aur dot products important hain.
- Neural networks aur optimization mein use hoti hai.

8. Convex Optimization

- Optimization ka matlab hai best solution dhoondhna.
- Gradient-based methods aur convex functions important hain.

9. Data Visualization

- Graphs, charts, scatter plots aur histograms se data samjha jata hai.
- Trends aur outliers identify karne mein helpful hota hai.

10. Hypothesis Function

- Hypothesis function model ka proposed relationship hota hai.
- Ye input data ko output se connect karta hai.

11. Hypothesis Testing

- Ye decide karta hai ki observed result statistically significant hai ya nahi.
- Null hypothesis aur alternate hypothesis important hain.

12. Data Distributions

- Normal distribution, uniform distribution aur binomial distribution important hain.
- Data ka behavior samajhne ke liye use hoti hain.

13. Data Preprocessing

- Missing values handle karna.
- Categorical data ko numeric mein convert karna.
- Feature scaling aur encoding.
- Train-test split.

14. Data Augmentation

- Data ko artificially increase karna.
- Image aur audio tasks mein kaafi useful hai.

15. Normalizing Data Sets

- Data ko same scale par lane ka process.
- Standardization aur normalization important hain.

16. Machine Learning Models

- Model ka matlab hai training data se learn karna aur new data par predict karna.
- Types: regression, classification, clustering, neural networks.

17. Supervised Learning

- Labeled data par train hota hai.
- Output known hota hai.
- Examples: classification aur regression.

18. Unsupervised Learning

- Labeled data bina train hota hai.
- Data ko groups ya patterns mein arrange karta hai.
- Examples: clustering, PCA.

Unit II – Neural Networks and Training

1. Neural Network Basics – Linearity vs Non-Linearity

- Linear models sirf straight-line relationship samjhate hain.
- Non-linear models complex patterns samajh sakte hain.

2. Activation Functions

- Sigmoid: probability-like output.
- ReLU: fast aur widely used.
- Tanh: zero-centered output.

3. Weights

- Weights inputs ko adjust karte hain.
- Ye model ki learning power ko define karte hain.

4. Bias

- Bias neuron ko shift karne mein help karta hai.
- Model ko better fit dene ke liye use hota hai.

5. Loss Function

- Loss function batata hai ki model kitna galat hai.
- Examples: MSE, Cross-Entropy.

6. Gradient Descent

- Ye sabse important topic hai.
- Model ke weights ko update karne ke liye use hota hai.
- Loss ko minimize karne ka process hai.

7. Multilayer Network

- Multiple layers ke through features learn kiye jate hain.
- Deep networks ke liye important hai.

8. Backpropagation

- Error ko backward propagate karke weights update karta hai.
- Neural network training ka core process hai.

9. Weight Initialization

- Initial weights ka selection training ko affect karta hai.
- Bad initialization se vanishing/exploding gradients aa sakte hain.

10. Training

- Model ko data par learn karaya jata hai.
- Epochs, batches aur iterations ka use hota hai.

11. Testing

- Model ko unseen data par test kiya jata hai.
- Generalization evaluate hoti hai.

12. Unstable Gradient Problem

- Gradients bahut chhote ya bade ho jaye to training weak ho jati hai.
- Solution: ReLU, Xavier/He initialization, batch normalization.

13. Autoencoders

- Input ko compress karke reconstruct karne ka model hai.
- Feature learning aur dimensionality reduction mein use hota hai.

14. Batch Normalization

- Training ko stable banata hai.
- Internal covariate shift ko reduce karta hai.

15. Dropout

- Overfitting ko reduce karta hai.
- Randomly neurons ko off kar diya jata hai.

16. L1 Regularization

- Sparsity laata hai.
- Feature selection mein helpful hai.

17. L2 Regularization

- Weights ko small rakhta hai.
- Overfitting ko reduce karta hai.

18. Momentum

- Gradient descent ko faster aur smoother banata hai.
- Oscillation ko kam karta hai.

19. Tuning Hyperparameters

- Learning rate, batch size, epochs aur layers ko tune karna.
- Model performance improve karne ke liye important hai.

Unit III – Convolutional Neural Networks (CNN)

1. CNN

- Images aur spatial data ke liye powerful model hai.
- Features automatically learn karta hai.
- Very important exam topic.

2. Flattening

- 2D feature maps ko 1D vector mein convert karna.

3. Subsampling

- Spatial dimensions ko reduce karna.
- Pooling se similar kaam hota hai.

4. Padding

- Output size ko preserve karne ke liye edges par zeros add karna.

5. Stride

- Filter kitni step mein move karta hai.
- Output size ko affect karta hai.

6. Convolution Layer

- Feature extraction ka basic layer hai.
- Filters input par apply hote hain.

7. Pooling Layer

- Spatial size reduce karta hai.

- Computation ko kam karta hai aur overfitting ko reduce karta hai.

8. Loss Layer

- Predicted aur actual output ke beech error compute karta hai.

9. Dense Layer

- Fully connected layer hoti hai.
- Final classification/regression ke liye use hoti hai.

10. 1x1 Convolution

- Channel dimension change karne ke liye use hoti hai.
- Network ko efficient banati hai.

11. Inception Network

- Multiple filter sizes ka use karta hai.
- Feature extraction ko better banata hai.

12. Input Channels

- RGB images mein channels 3 hote hain.
- Channel concept important hai.

13. Transfer Learning

- Ek trained model ko dusre task ke liye reuse karna.
- Time aur data save karta hai.

14. One Shot Learning

- Bahut kam examples se learning karna.
- Face recognition mein use hota hai.

15. Dimension Reduction

- Features ko reduce karke model ko simple banana.
- PCA aur autoencoders ka use ho sakta hai.

16. Implementation of CNN

- CNN ko practical form mein implement karna.
- TensorFlow, Keras aur PyTorch common tools hain.

Unit IV – RNN, NLP and Reinforcement Learning

1. Recurrent Neural Network (RNN)

- Sequential data ke liye use hoti hai.
- Previous information ko remember kar sakti hai.

2. LSTM

- RNN ke vanishing gradient problem ko solve karta hai.
- Long-term dependencies ko remember kar sakta hai.

3. GRU

- LSTM ka simpler version hai.
- Less parameters aur faster training.

4. Translation

- Source language ko target language mein convert karna.
- Sequence-to-sequence models mein use hota hai.

5. Beam Search

- Best sequence generate karne ka method hai.
- NLP mein translation aur text generation mein use hota hai.

6. Beam Width

- Kitne candidate sequences ko consider karna hai.
- Beam width bada hone par search expensive hota hai.

7. BLEU Score

- Machine translation quality evaluate karne ka metric hai.
- Higher BLEU better quality indicate karta hai.

8. Attention Model

- Important words par focus karta hai.
- Sequence models mein very useful hai.

9. Reinforcement Learning

- Agent environment ke saath interact karta hai aur reward seek karta hai.
- Goal: maximum cumulative reward.

10. RL Framework

- Agent, environment, state, action aur reward ka structure.

11. MDP

- Decision-making framework hai.
- State transition aur reward model par based hota hai.

12. Bellman Equations

- Value function ko recursively define karte hain.
- Dynamic programming ke liye important hai.

13. Value Iteration

- Optimal policy dhoondhne ka iterative method hai.

14. Policy Iteration

- Policy ko improve karne ka method hai.

15. Actor-Critic Model

- Policy aur value function dono ko learn karta hai.
- RL mein popular approach hai.

16. Q-Learning

- Model-free reinforcement learning method hai.
- Agent learns action-value function.

17. SARSA

- On-policy RL algorithm hai.
- Q-learning ke comparison mein thoda alag hai.

Unit V – Applications and Advanced Topics

1. SVM

- Classification aur regression dono ke liye use hota hai.
- Maximum margin boundary dhoondhta hai.

2. Bayesian Learning

- Probabilistic reasoning par based hai.
- Bayes theorem aur posterior probability important hain.

3. Application of ML in Computer Vision

- Image classification, object detection aur face recognition mein use hoti hai.

4. Application of ML in Speech Processing

- Speech recognition, speaker identification aur audio analysis mein use hoti hai.

5. Application of ML in NLP

- Text classification, sentiment analysis, machine translation aur chatbots mein use hoti hai.

6. Case Study: ImageNet Competition

- Deep learning aur CNNs ki progress ko dikhane wala famous case study hai.

Extra Revision Points

- Machine Learning ka end-to-end workflow: Data Collection -> Data Preprocessing -> Training -> Validation -> Testing -> Deployment.
- Overfitting vs Underfitting:
- Overfitting: model training data ko bahut achhe se yaad kar leta hai, par new data par weak hota hai.
- Underfitting: model simple hone ki wajah se patterns samajh nahi pata.
- Bias-Variance Tradeoff:
- High bias => underfitting.
- High variance => overfitting.
- Feature Engineering:
- Important features ko select aur create karna model performance improve karta hai.
- Evaluation Metrics:
- Classification ke liye Accuracy, Precision, Recall, F1-Score.
- Regression ke liye MSE, RMSE, MAE.
- Confusion Matrix basics:
- TP, TN, FP, FN ko samjho.
- Precision aur Recall ka relation samjho.
- Deep Learning ke main advantages:
- Automatic feature extraction.
- Large datasets par strong performance.
- Complex patterns samajhne mein powerful.

Important Formulas

1. Regression and Classification

- Linear Regression: $y = wx + b$
- $MSE = (1/n) \sum (y_i - \hat{y}_i)^2$
- $RMSE = \sqrt{MSE}$
- $MAE = (1/n) \sum |y_i - \hat{y}_i|$
- Logistic Regression Probability: $p = 1 / (1 + e^{-z})$, where $z = wx + b$

2. Probability and Bayes

- Conditional Probability: $P(A|B) = P(A \cap B) / P(B)$
- Bayes Theorem: $P(A|B) = [P(B|A) P(A)] / P(B)$
- Joint Probability: $P(A \cap B) = P(A) P(B|A)$

3. Statistics

- Mean: $\mu = (1/n) \sum x_i$
- Variance: $\sigma^2 = (1/n) \sum (x_i - \mu)^2$

- Standard Deviation: $\sigma = \sqrt{\sigma^2}$
- Correlation: $r = \text{Cov}(x,y) / (\sigma_x \sigma_y)$

4. Neural Networks

- Gradient Descent: $w_{\text{new}} = w_{\text{old}} - \eta \partial L / \partial w$
- Bias Update: $b_{\text{new}} = b_{\text{old}} - \eta \partial L / \partial b$
- Sigmoid Activation: $\sigma(x) = 1 / (1 + e^{-x})$
- Tanh Activation: $\tanh(x) = (e^x - e^{-x}) / (e^x + e^{-x})$
- ReLU Activation: $\text{ReLU}(x) = \max(0, x)$
- Cross Entropy Loss: $L = - \sum y \log \hat{y}$

5. CNN

- Output Size: $O = \text{floor}((W - F + 2P) / S) + 1$

6. Classification Metrics

- Accuracy = $(TP + TN) / (TP + TN + FP + FN)$
- Precision = $TP / (TP + FP)$
- Recall = $TP / (TP + FN)$
- F1 Score = $2PR / (P + R)$

7. SVM and RL

- SVM Decision Function: $f(x) = w \cdot x + b$
- Q-learning Update: $Q(s,a) = Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$

Simple Diagrams

1. Machine Learning Workflow

Data Collection -> Data Preprocessing -> Feature Engineering
-> Training -> Validation -> Testing -> Deployment

2. Simple Neural Network

Input Layer -> Hidden Layer -> Output Layer
 $x_1, x_2, x_3 \rightarrow h_1, h_2, h_3 \rightarrow y$

3. CNN Flow

Input Image -> Convolution -> Activation -> Pooling -> Flatten -> Dense Layer -> Output

Important Exam Points

- High priority topics: Gradient Descent, Backpropagation, CNN, LSTM, MDP, Q-Learning.
- Medium priority topics: Regression, SVM, Bayesian Learning, Transfer Learning, Attention Model.
- Low priority topics: Hypothesis Testing, Beam Search, One-Shot Learning, Weight Initialization.
- Definitions, differences, formulas aur applications yaad rakho.

- Viva/exam ke liye algorithms ka comparison aur use cases clearly samjho.